

Week 1

**Implement "Tic-tac-Toe" in general without using any standard algorithm.**

```
board = {str(i): str(i) for i in range(1, 10)}
```

```
def print_board():
```

```
    print("\n" + board["1"] + " | " + board["2"] + " | " + board["3"])
    print("-----")
    print(board["4"] + " | " + board["5"] + " | " + board["6"])
    print("-----")
    print(board["7"] + " | " + board["8"] + " | " + board["9"] + "\n")
```

```
def check_win(player):
```

```
    win_conditions = [
        ["1", "2", "3"], ["4", "5", "6"], ["7", "8", "9"],
        ["1", "4", "7"], ["2", "5", "8"], ["3", "6", "9"],
        ["1", "5", "9"], ["3", "5", "7"]
    ]
    for condition in win_conditions:
        if board[condition[0]] == board[condition[1]] == board[condition[2]] == player:
            return True
    return False
```

```
def check_draw():
```

```
    return all(value in ["X", "O"] for value in board.values())
```

```
def play_game():
```

```
    current_player = "X"
```

```
    while True:
```

```
        print_board()
```

```
        move = input(f"Player {current_player}, choose a spot (1-9): ")
```

```
        if move in board and board[move] not in ["X", "O"]:
```

```
            board[move] = current_player
```

```
            if check_win(current_player):
```

```
                print_board()
```

```
    print(f"Player {current_player} wins!")  
    break
```

```
if check_draw():  
    print_board()  
    print("It's a draw!")  
    break
```

```
    current_player = "O" if current_player == "X" else "X"  
else:  
    print("Invalid move or spot already taken. Try again.")
```

```
if __name__ == "__main__":  
    play_game()
```

## Output

```
board = {str(i): str(i) for i in range(1, 10)}

def print_board():

    print("\n" + board["1"] + " | " + board["2"] + " | " + board["3"])
    print("-----")
    print(board["4"] + " | " + board["5"] + " | " + board["6"])
    print("-----")
    print(board["7"] + " | " + board["8"] + " | " + board["9"] + "\n")

def check_win(player):

    win_conditions = [
        ["1", "2", "3"], ["4", "5", "6"], ["7", "8", "9"],
        ["1", "4", "7"], ["2", "5", "8"], ["3", "6", "9"],
        ["1", "5", "9"], ["3", "5", "7"]
    ]
    for condition in win_conditions:
        if board[condition[0]] == board[condition[1]] == board[condition[2]] == player:
            return True
    return False

def check_draw():

    return all(value in ["X", "O"] for value in board.values())

def play_game():
    current_player = "X"

    while True:
        print_board()
        move = input(f"Player {current_player}, choose a spot (1-9): ")

        if move in board and board[move] not in ["X", "O"]:
            board[move] = current_player

            if check_win(current_player):
                print_board()
                print(f"Player {current_player} wins!")
                break

            if check_draw():
                print_board()
                print("It's a draw!")
                break

            current_player = "O" if current_player == "X" else "X"
        else:
            print("Invalid move or spot already taken. Try again.")
```

```

... 1 | 0 | X
-----
4 | 5 | 6
-----
7 | 8 | 9

Player X, choose a spot (1-9): 4

1 | 0 | X
-----
X | 5 | 6
-----
7 | 8 | 9

Player O, choose a spot (1-9): 5

1 | 0 | X
-----
X | 0 | 6
-----
7 | 8 | 9

Player X, choose a spot (1-9): 9

1 | 0 | X
-----
X | 0 | 6
-----
7 | 8 | X

Player O, choose a spot (1-9): 6

1 | 0 | X
-----
X | 0 | 0
-----
7 | 8 | X

Player X, choose a spot (1-9): 1

X | 0 | X
-----
X | 0 | 0
-----
7 | 8 | X

Player O, choose a spot (1-9): 8

X | 0 | X
-----
X | 0 | 0
-----
7 | 0 | X

Player O wins!

```